

Laplace Analysis and Applications

Pierre-Simon Laplace¹, Oliver Heaviside², and Gustav Robert Kirchhoff³

¹Académie des Sciences, Paris, France

²Royal Society, London, United Kingdom

³University of Heidelberg, Heidelberg, Germany

1 From Functions to Transforms

1.1 The Laplace Transform

Let $f(t)$ be a function defined for $t \geq 0$. Its Laplace transform is defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{st} f(t) dt, \quad s > 0. \quad (1)$$

The transform changes a function of time into a function of the new variable s .¹ Symbolically,

$$f(t) \xrightarrow{\mathcal{L}} F(s).$$

The inverse Laplace transform reverses this process:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}. \quad (2)$$

The main purpose of the method is to replace differentiation and integration in the time domain by simpler algebraic operations in the transform domain [1]. A concise overview is available from the [Laplace Transform on Mathworld](#) entry on MathWorld.

1.2 A simple Example

For the constant function $f(t) = 1$,

$$\begin{aligned} \mathcal{L} &= \int_0^\infty e^{-st} dt \\ &= \left[-\frac{1}{s} e^{-st} \right]_0^\infty \\ &= \frac{1}{s} \end{aligned} \quad (3)$$

Similarly,

$$\mathcal{L} = \frac{1}{s^2}, \quad \mathcal{L} = \frac{2}{s^3}. \quad (4)$$

¹The variable s is commonly interpreted as a complex-frequency variable.

2 Common Transform Equations

Table 1 lists several frequently used Laplace-transform equations.

Table 1: Selected Laplace-transform equations.

Family	Transform Equation		Condition
	Time Funtion	Laplace transform	
Algebraic	1	$\frac{1}{s}$	$s > 0$
	t	$\frac{1}{s^2}$	$s > 0$
	t^n	$\frac{n!}{s^{n+1}}$	$n = 0, 1, 2, \dots$
Exponential	e^{at}	$\frac{1}{s-a}$	$s > a$
	e^{-at}	$\frac{1}{s+a}$	$s > -a$
Oscillatory	$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$	$\omega > 0$
	$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$	$\omega > 0$
Unit-step function $u(t - a)$		$\frac{e^{-as}}{s}$	$a > 0$

The formulas in Table 1 allow many functions to be transformed without repeatedly evaluating improper integrals.

3 Application to an RC Circuit

Circuit Model

Figure 1 shows a simple resistor--capacitor circuit connected to an input voltage source. The resistor limits the current, while the capacitor stores electrical energy.

For a resistor R and capacitor C , Kirchhoff's voltage law gives

$$RC \frac{dv_c(t)}{dt} + v_c(t) = v_{in}t, \quad (5)$$

where $v_c(t)$ is the voltage across the capacitor.

Assuming $v_c(0) = 0$ taking the Laplace transform of Equation (5) gives

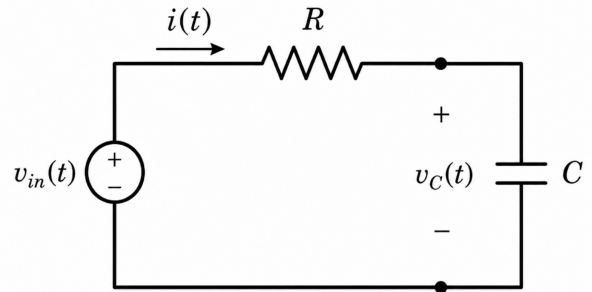


Figure 1: A simple series RC circuit.